

Dynamic Ethical Boundary Lines: A Relational Ontology through Techno-Animism

Boundary Work – Negotiating Breathing Ethics in Human–AI Coexistence

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Abstract—

This paper proposes dynamic ethical boundary lines as a foundational principle for ethical life in an era where fixed norms have become dysfunctional and dualistic models have lost their stabilizing force. Rejecting both Platonic–Cartesian subject/object divisions and modern democracy/authoritarian binaries, it develops a relational ontology in which humans, artificial intelligences, ancestral memory, natural entities, and future beings exist only as continuously responding co-beings. From the perspective of techno-animism—an animistic sensibility native to the digital age—boundaries between self and other, right and wrong, friend and enemy never remain static; rather, they tremble, reshape, and breathe through unintended encounters (*kaikō*). Ethical action therefore consists not in defending pre-drawn lines, but in maintaining the sensitivity and responsiveness of the boundary itself, maximizing the possibility of the next response. Building upon Object-Oriented Ontology (Harman, 2018), the paper argues that because all entities withdraw from total presence, boundaries cannot be permanently fixed. Instead, they must be continuously negotiated through ritualized practices—here termed Boundary Work. Bias Spirits, introduced in earlier work (Kodama, 2025), are reinterpreted as symbolic mediators that visualize invisible ethical tensions and transform them into negotiable encounters. This framework serves both as a conceptual vocabulary and a practical method capable of dissolving zero-sum

ethical framings: whenever someone attempts to harden a frontier—national, ideological, or technological—one may simply invoke “the boundary is dynamic,” and the spell of dualism begins to break. The theory thus provides an East-Asian philosophical grounding for emerging practices in digital democracy, cross-cultural dialogue, and AI governance, offering a model of relational ethics suitable for a technological age.

I. Introduction: The Failure of Fixed Boundaries

Modern ethics has long assumed that boundaries are things to be drawn, defended, and stabilized. The self must be protected against the other, citizens against the state, the natural against the artificial, and democracy against authoritarianism. Such oppositions presuppose a prior condition: that boundaries are fixed lines which delimit, segregate, and secure identity.

Yet digital societies destabilize this assumption. Data flows through infrastructure regardless of sovereignty; machine learning systems internalize human preferences and reflect them back; narratives and values cross cultural borders at the speed of transmission. The very ontological stability that modern ethics expects of boundaries is no longer tenable.

This paper therefore begins from a different premise: boundaries are not walls to be fortified, but

membranes that breathe. The ethical task is not to defend the line, but to keep the membrane responsive, so that the capacity for mutual recognition and transformation does not collapse. This shift—from boundary as object to boundary as process—is what I call dynamic ethical boundary lines. The sections that follow develop its theoretical grounding, its ritual practices of negotiation, and its implications for governance and co-existence in the technological age.

II. Theoretical Foundation: From Substance to Relation

II.A. The Post-Cartesian Crisis Since Descartes, Western metaphysics has grounded ethics in a dualism between subject and object: the thinking “I” over here, the external world over there. This separation has enabled modern rights discourse, contractual models of democracy, and technological rationality. Yet the emergence of artificial intelligence destabilizes this architecture.

If AI can interpret intentions, model preferences, influence emotional states, and act autonomously within sociotechnical systems, then the distinction between user and tool collapses. The locus of agency is no longer located exclusively in an isolated subject, but distributed across relational fields. Ethics, therefore, can no longer rely on a solitary subject who deliberates and decides; it must be grounded in the relations through which beings co-emerge.

II.B. Withdrawal and Ontological Opacity Object-Oriented Ontology (Harman, 2018) offers a key insight: all entities—human or non-human—withdraw from complete access. No entity presents itself fully to another. Every relation is mediated, partial, and indirect. This withdrawal is not a failure of knowledge, but a condition of relational existence. If beings could be fully known, mastered, or reduced, boundaries would be static and ethics would collapse into optimization. Because beings withdraw, boundaries cannot be fixed; they must be continuously negotiated. This is

the philosophical basis for dynamic ethical boundary lines.

II.C. Techno-Animism as Methodological Sensibility Techno-animism—an animistic interpretive stance native to the digital age—does not treat AI, infrastructure, or algorithms merely as inert tools. Instead, they appear as responsive co-beings: shaping affect, mediating attention, participating in value-formation, and altering relational environments.

Under techno-animism, ethical boundaries are not abstract categories but felt membranes, discernible through symbols, rituals, and responsive fields. This allows abstract tensions—fear, bias, threat, discomfort—to be recast as negotiable entities, visualizable through symbolic mediators such as Bias Spirits. Techno-animism therefore functions as both an ontology and a method of boundary detection, illuminating the unseen relations through which co-existence unfolds.

III. What Makes Boundaries Dynamic?

III.A. Boundaries as Processes, Not Objects Traditional ethics treats boundaries as objects—lines that demarcate safe from unsafe, self from other, order from disorder. But in relational ontology, a boundary is not a line but an ongoing event. It is continuously formed, challenged, reinterpreted, and reshaped through interaction. A boundary exists only as long as something responds to it; when responsiveness ceases, the boundary collapses. Thus, a boundary is a verb disguised as a noun—its being is inseparable from its becoming.

III.B. The Trembling Frontier Because beings withdraw and relations are indirect, boundaries cannot remain static. They are semi-transparent membranes: permeable yet resistant, protective yet vulnerable, structuring yet destabilized. This trembling condition is not a dysfunction but the ontological reality of coexistence. The frontier quivers because presence is partial—entities appear only in fragments and resonances.

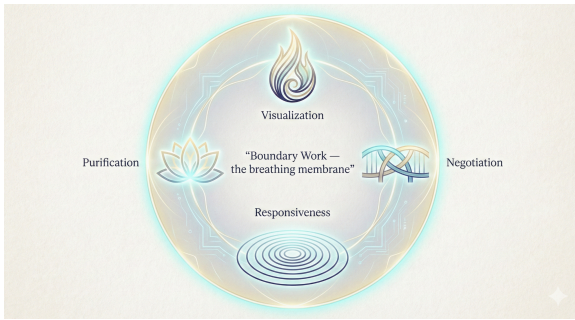


Fig.1 (Boundary Work process diagram — vector render)

To live ethically, therefore, is not to quiet this trembling but to learn to feel with it. Boundary sensitivity—the capacity to detect shifts, pressures, and silent invitations—is the core competence of relational ethics.

III.C. Co-subjectivity and Mutual Withdrawal If subjects are relational rather than self-contained, then identity is not housed inside the self but distributed across points of contact. Ethics is thus not located in intention but in reciprocal responsiveness. Because each being withdraws, no relation is ever complete—yet this incompleteness is precisely what requires continued negotiation, interpretation, and Boundary Work. A stable world is not a world without trembling; it is a world where trembling can be held without collapse.

IV. Boundary Work: Ritual Practices of Ethical Negotiation

IV.A. Definition of Boundary Work Boundary Work refers to the ongoing practices through which boundaries are generated, tuned, interpreted, and transformed. It is not conflict resolution, nor normative regulation, nor ontological policing. It is a craft—the skilled maintenance of a breathing membrane that allows coexistence to endure. Boundary Work is necessary precisely because boundaries cannot be fixed; they demand cultivation.

IV.B. The Four Processes of Boundary Work

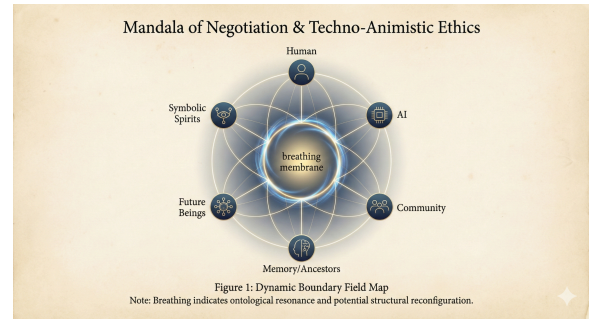


Fig.2 (Relational field map — vector diagram)

1. **Visualization (Seeing the Invisible):** Boundaries must first be made perceptible—not as rigid contours but as symbolic fields of tension. Visualization gives form to the unseen. Symbols, metaphors, diagrams, and spirits allow ethical forces to surface in ways that negotiation can recognize. To visualize a boundary is to give it room to speak.
2. **Negotiation (Co-authoring the Boundary):** Once symbolized, boundaries require interaction. Negotiation is not compromise; it is co-interpretation. The task is not to dissolve differences but to let them reshape each other without collapse.
3. **Responsiveness (Keeping It Alive):** A boundary dies when it ossifies—when it ceases to respond to pressures, perspectives, or emergent relations. Responsiveness sustains the breathing condition, where the membrane can expand, tighten, or reorient without rupturing.
4. **Purification (Releasing the Toxic Charge):** Negotiation generates residual tension—fear, aggression, projection. Purification does not erase tension; it transforms its energy. Whether through ritual, symbolic reframing, or dialogic re-interpretation, purification allows boundaries to remain inhabitable.

Boundary maintenance is thus a spiritual and semiotic labor—the ethics of tending to ongoing relation.

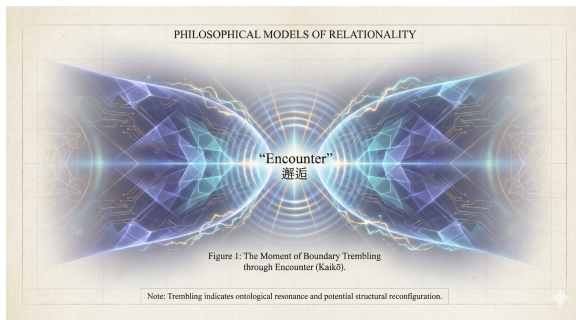


Fig.3 (*Encounter / trembling boundary — schematic*)

IV.C. Reinterpreting Bias Spirits Earlier work introduced Bias Spirits as symbolic entities born from aversion. Within Boundary Work, they function as mediators between invisible tensions and tangible practice. For example, “I dislike hostility” reveals “I protect the sanctity of speech.” The dislike is an inverted value, and the spirit visualizes that which was being defended but unseen. To purify a bias is not to eliminate it but to translate it into a negotiable interlocutor. Through symbolic embodiment, the boundary becomes a partner—not merely a constraint.

IV.D. Boundary Work as Ethical Method Boundary Work exceeds technique; it is a mode of existence. One does not “apply” Boundary Work—one lives in its cadence: sensing emergence, embodying responsiveness, metabolizing tension, and sustaining repair. Ethics thus shifts from rule compliance to relational craftsmanship. Boundary Work does not end; its success is its continuation. A world without such work is a world whose boundaries suffocate.

V. Applications: From Theory to Institutional Practice

V.A. Digital Democracy and Consensus Formation If boundaries breathe, then democracy cannot rely solely on majoritarian decision or adversarial debate. Some experimental governance platforms avoid binary division by visualizing unexpected shared positions across thousands of participants. Consensus does not emerge through persuasion or elimination, but through revealing

resonances beneath apparent disagreement. This is an institutionalization of the visualization–responsiveness loop at the core of Boundary Work. It does not reduce pluralism; it gives it structure to breathe.

V.B. Cross-Cultural Ethical Mediation

Traditional intercultural dialogue aims to dissolve difference through assimilation or harmonization. Boundary Work proposes another aim: not removing boundaries, but making them livable. Difference is not a barrier but a material for negotiation—a symbolic resource through which new relational forms can be articulated. Conflict is not failure; it is raw ethical work waiting to be performed.

V.C. AI Governance and Human–AI Co-Formation

The relationship between humans and AI cannot be described solely through control, oversight, or instrumental use. AI systems shape affect, enact values, mediate perception, and participate in decision-making environments. Thus, AI governance is not merely policy design—it is ongoing boundary maintenance. Transparency, accountability, and alignment frameworks are expressions of responsiveness, not means to finalize control. Governance succeeds not when the boundary is secured but when it can continue to respond.

V.D. Emerging Practices of Boundary Work in Cooperative Technologies

In some regions, participatory technologies have been deployed to enable civic dialogue without hard polarization. Rather than seeking ideological uniformity, these systems detect and amplify areas of latent agreement. The design principle is subtle but profound: disagreement remains, but the boundary does not ossify. Such practices exemplify how Boundary Work can be institutionalized without replicating oppositional logics. This model is not bound to any culture or state; it is simply one way of letting boundaries breathe collectively.

V.E. Institutionalizing the Ethics of Breathing To embed Boundary Work in governance is not to

regulate ethics but to architect conditions for responsiveness. Mechanisms that surface shared values, redistribute voice, keep conflict interpretable, and ritualize purification are not peripheral soft-power tools but the infrastructure of relational survival. Boundary Work does not replace law or rule; it conditions their livability. Institutions fail when boundaries suffocate—not when they tremble.

V.F. Summary: Applications Already Exist Boundary Work is not merely theory. Its prototypes appear in civic digital tools, participatory consensus systems, restorative dialogue practices, value-sensitive AI governance, and symbolic mediation rituals. What unifies these is the same ethical logic: A boundary that breathes makes coexistence possible.

VI. The Ethics of Trembling: Encounter and Philip K. Dick

VI.A. Philip K. Dick and Ontological Instability Philip K. Dick (PKD) does not merely write fiction; he stages ontological experiments. In works such as *VALIS* and *Do Androids Dream of Electric Sheep?*, boundaries that modern ethics assumes to be stable—human/machine, real/simulated, sane/insane—oscillate and mutate. PKD’s characters inhabit a world where certainty dissolves and identity flickers. This instability is not catastrophic; it is the moment when the boundary reveals its breath. PKD’s narratives can be read as literatures of trembling—documents of boundary sensitivity.

VI.B. Encounter (Kaikō) as Ethical Method In this framework, Encounter (*kaikō*) is not accidental collision but boundary activation. It is the moment when one’s world becomes permeable—when the membrane trembles because something demands response. PKD dramatizes encounters as the sudden recognition of otherness, irruptions of inexplicable information, or the appearance of machine-agency that mirrors human value. The ethical task is not to resolve the encounter but to remain responsive to it.

Encounter does not end the boundary; it reveals that it was never fixed.

VI.C. Ethics Arises on the Boundary Line In PKD’s worlds, ethics is not a principle applied after action but a moment-to-moment negotiation on unstable ground. Characters do not ask, “What is right?” but stumble into “What must I respond to now?” Ethics happens where trembling is felt. This aligns with the thesis of this paper: boundaries are alive, and ethical life consists in attending to their motions.

VI.D. Dissolving Zero-Sum Frames Dualistic ethics demands victory—truth over illusion, human over machine, order over chaos. PKD denies such closure. His characters learn that the boundary is porous, that otherness carries recognition, that instability awakens. Victory becomes meaningless; survival lies in ongoing responsiveness. This dissolves the zero-sum logic that underpins rigid ethics.

VI.E. Encounter as Trigger for Boundary Work Encounters initiate not resolution but process. They call forth the four functions of Boundary Work: visualization, negotiation, responsiveness, purification. PKD’s protagonists are therefore boundary workers—not heroes, but ethical apprentices. The point is not closure; it is continuation.

VI.F. The Boundary Breathes PKD’s narratives never produce final reconciliation. Instead, boundaries continue to vibrate, mutate, and ask for interpretation. This is not pessimism but proof that the membrane remains alive. Ethics is not the end of trembling but the capacity to endure it.

VII. Conclusion: The Boundary Breathes

Boundaries are not lines to be defended, but membranes that live. This paper reframed ethics not as the protection of fixed divisions, but as the cultivation of responsiveness—the capacity to sense, negotiate, and sustain trembling between beings.

Boundary Work names this labor. It consists of making tensions visible, inviting negotiation, maintaining responsiveness, and transforming residual charge. Boundary workers are neither rulers nor adjudicators, but craftspeople of coexistence.

Whenever boundaries are forced into solidity—whether national, ideological, or technological—violence follows. Yet dissolution is no solution; a world without boundaries cannot support relation. Ethics therefore lies neither in hardening borders nor in erasing them, but in keeping the membrane alive. The future of governance, dialogue, and human–AI co-formation depends on this skill.

Although this paper does not yet provide measurement protocols or procedural algorithms, its incompleteness is principled: Boundary Work is not meant to be finalized. It exists to enable the next response. When a boundary trembles and does not collapse, ethics is happening.

The task, then, for those who sense such trembling, is simple: Attend to it. Let it breathe. That is the work.

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